

Intro to Web Development - Assignment Answers

1. Difference between Frontend, Backend, and Full-Stack Development

Frontend development focuses on the visual aspects of a website - everything a user interacts with directly. It involves technologies like HTML, CSS, and JavaScript. Example: the layout and design of Amazon's product page.

Backend development deals with server-side logic, databases, and authentication. It ensures the data flow between the frontend and the database is smooth. Example: when you log in to Instagram, the backend verifies your credentials.

Full-stack developers handle both frontend and backend parts. They can build complete web applications end-to-end. Example: a full-stack developer can create both the interface and the database logic for an online store.

2. Client-Server Model Diagram

The client-server model works as follows:

Client (browser) -> Internet -> Web Server -> Database -> Response back to Client.

Example: When you open Facebook, your browser (client) requests data from Facebook's server, which fetches user data from a database and sends it back.

3. How a Browser Requests and Displays a Web Page

When you enter a URL, the browser sends an HTTP request to the server. The server processes it and sends back the requested files (HTML, CSS, JS). The browser then renders these files into a visual web page using its rendering engine.

4. Tools Required for Web Development

1. Code Editor (VS Code) - for writing code.
2. Browser (Chrome/Firefox) - for testing.
3. Version Control (Git) - for managing versions.
4. Terminal - to run commands.
5. Node.js - to run JavaScript outside browsers.

5. What is a Web Server?

A web server hosts and serves web pages to clients via HTTP/HTTPS protocols. Examples include Apache, Nginx, and Microsoft IIS.

6. Roles in a Web Project

Frontend Developer - designs UI/UX and writes HTML, CSS, JS.

Backend Developer - manages logic, APIs, and server operations.

Database Administrator - maintains, secures, and optimizes the database.

7. VS Code Installation and Configuration

Install VS Code from the official website. Then add extensions for HTML, CSS, JavaScript, and Live Server. You can open a folder, create files (index.html, style.css, script.js), and start coding.

8. Static vs Dynamic Websites

Static websites show the same content to all users (e.g., portfolio sites). Dynamic websites generate content based on user input or data (e.g., YouTube, Facebook).

9. Web Browsers and Rendering Engines

1. Google Chrome - Blink
2. Mozilla Firefox - Gecko
3. Safari - WebKit
4. Microsoft Edge - Blink

5. Opera - Blink

Each engine interprets and displays HTML/CSS/JS differently, causing slight variations in rendering.

10. Basic Web Architecture Flow

Client -> Server -> Database -> API -> Response -> Client.

The client sends a request, the server processes it, fetches data from the database through APIs, and returns the response.